

SUPPLEMENTARY MATERIAL

CLASSIFYING INTEGRAL GROTHENDIECK RINGS UP TO RANK 5 AND BEYOND

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I. INTEGRAL DRINFELD RINGS

I.1. Up to rank 5. This section presents the comprehensive list of integral Drinfeld rings up to rank 5—including their global FPdim, type, duality, formal codegrees, and fusion data. Copy-pastable data can be found in the file `GeneralUpToRank5DataOnly.txt`, located in the `Data/General` directory of [8]. For each case, either an explicit categorification is provided, or a reference is given to a theoretical result ruling out its existence.

I.1.1. *Rank 1.* Trivial case

I.1.2. *Rank 2.*

(1) FPdim 2, type $[1, 1]$, duality $[0, 1]$, fusion data:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Formal codegrees: $[2, 2]$,
- Property: simple,
- Categorification status: $\text{Rep}(C_2)$.

I.1.3. *Rank 3.*

(1) FPdim 3, type $[1, 1, 1]$, duality $[0, 2, 1]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- Formal codegrees: $[3, 3, 3]$,
- Property: simple,
- Categorification status: $\text{Rep}(C_3)$.

(2) FPdim 6, type $[1, 1, 2]$, duality $[0, 1, 2]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- [illegible]

(3) FPdim 2860, type $[1, 11, 12, 12, 15, 25, 40]$, duality $[0, 1, 2, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 6 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 1 & 2 & 3 & 3 & 4 & 6 & 8 \\ 0 & 4 & 4 & 4 & 4 & 8 & 15 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & 2 & 2 & 2 & 4 & 6 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 10 \\ 0 & 4 & 4 & 4 & 4 & 8 & 15 \\ 1 & 6 & 7 & 7 & 10 & 15 & 21 \end{bmatrix}$$

- Formal codegrees: $[4, 4, 5, 5, 13, 44, 2860]$,
- Property: simple, non-1-Frobenius,
- Categorification status: open, non-braided.

(4) FPdim 3192, type $[1, 15, 16, 17, 18, 24, 39]$, duality $[0, 1, 2, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 1 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 2 & 2 & 2 & 3 & 3 & 6 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 3 & 3 & 3 & 2 & 1 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 3 & 3 & 4 & 4 & 7 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \\ 1 & 7 & 8 & 9 & 8 & 12 & 18 \end{bmatrix}$$

- Formal codegrees: $[3, 3, 6, 8, 42, 57, 3192]$,
- Property: simple, non-1-Frobenius, non-3-positive
- Categorification status: open, non-braided, non-unitary

(5) FPdim 4284, type $[1, 17, 17, 20, 28, 35, 36]$, duality $[0, 2, 1, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 3 \\ 0 & 1 & 2 & 1 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 2 & 2 & 3 \\ 0 & 2 & 1 & 2 & 2 & 2 & 2 \\ 0 & 2 & 1 & 1 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 2 & 3 & 3 \\ 0 & 2 & 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 & 3 & 2 & 3 \\ 0 & 2 & 2 & 2 & 6 & 4 & 4 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 3 & 3 & 4 & 4 & 6 & 6 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 7 & 8 & 8 \\ 0 & 4 & 4 & 4 & 8 & 8 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 7 & 8 & 8 \\ 1 & 5 & 5 & 6 & 8 & 10 & 10 \\ 0 & 5 & 5 & 6 & 8 & 10 & 11 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \\ 0 & 3 & 3 & 4 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 8 & 8 & 8 \\ 0 & 5 & 5 & 6 & 8 & 10 & 11 \\ 1 & 5 & 5 & 6 & 8 & 10 & 11 \end{bmatrix}$$

- Formal codegrees: $[3, 4, 7, 7, 9, 51, 4284]$,
- Property: simple, non-1-Frobenius, non-3-positive
- Categorification status: open, non-braided, non-unitary

These fusion rings also invite questions in the spirit of [1, Question 1.13].

I.4. Ranks 8 and 9. There are 792 integral 1-Frobenius Drinfeld rings of rank 8 with $\text{FPdim} \leq 25000$, and 1292 such rings of rank 9 with $\text{FPdim} \leq 2000$. Copy-pastable data can be found in the files `1FrobR8d25000.txt` and `1FrobR9d2000.txt`, located in the `Data/General` directory of [8].

I.5. Addressing divisibility. The classification at rank r begins with the list of length r Egyptian fractions representing the possible

$$\sum_V \sum_{i=1}^{n_V^2} \frac{1}{n_V f_V} = 1,$$

where the (f_V) are the formal codegrees of the Drinfeld ring; see [1, §2.2]. Under the Drinfeld assumption, each formal codegree f_V divides f_1 . Moreover, the Drinfeld ring is commutative if and only if $n_V = 1$ for all V . In this case, we can restrict to Egyptian fractions of length r satisfying the divisibility condition $f_V \mid f_1$ for all V .

In the noncommutative setting, it may happen that $n_V f_V$ does not divide f_1 for some V . However, as verified in [1, §III], no such exceptions occur up to rank 8 (but they do appear at rank 9; see [1, §2.2]). Therefore, for ranks $r \leq 8$, it is safe to restrict to Egyptian fractions of length r that satisfy the divisibility condition.

At rank 9, [1, Lemma 9.3] shows that the complexified noncommutative Drinfeld ring must be isomorphic to either $\mathbb{C} \oplus M_2(\mathbb{C})^2$ or $\mathbb{C}^5 \oplus M_2(\mathbb{C})$. The exception corresponds to Egyptian fractions of length 5 or 7 with one or two terms having $n_V = 2$, and at least one violating the divisibility condition $n_V f_V \mid f_1$.

We verified that for $\text{FPdim} \leq 32000$, the values arising from these exceptional cases are already covered by those for the Egyptian fractions of length 9 satisfying the divisibility condition. Details of this computation can be found in the file `InvestNCRank9Except.txt` of the `Data/EgyptianFractionsDiv/Except` folder of [8].

II. MNSD DRINFELD RINGS

As established in [1, §8], the Grothendieck ring of any odd-dimensional integral fusion category over $\mathbb{C}$ is an MNSD integral Drinfeld ring ([1, Definition 8.6]). This section provides a complete classification of such rings up to rank 9.

II.1. Up to rank 5. There are four MNSD integral Drinfeld rings up to rank 5, contained in §I.1, namely the Grothendieck rings of $\text{Rep}(G)$, with $G = C_1, C_3, C_5, C_7 \rtimes C_3$.

II.2. **Rank 7.** Here is the complete list of 4 MNSD integral Drinfeld ring of rank 7. Copy-pastable data can be found in the file `MNSDRank7DataOnly.txt.txt`, located in the `Data/Odd` directory of [8].

- (1) FPdim 7, type $[1, 1, 1, 1, 1, 1, 1]$, duality $[0, 6, 5, 4, 3, 2, 1]$, fusion data: see §I.3.1.
 (2) FPdim 39, type $[1, 1, 1, 3, 3, 3, 3]$, duality $[0, 2, 1, 6, 5, 4, 3]$, fusion data:

[illegible]

- Formal codegrees: $[3, 3, 13, 13, 13, 13, 39]$.
- Categorification status: $\text{Rep}(C_{13} \rtimes C_3)$.

- (3) FPdim 55, type $[1, 1, 1, 1, 1, 5, 5]$, duality $[0, 4, 3, 2, 1, 6, 5]$, fusion data:

[illegible]

- Formal codegrees: $[5, 5, 5, 5, 11, 11, 55]$,
- Categorification status: $\text{Rep}(C_{11} \rtimes C_5)$.

- (4) FPdim 903, type $[1, 1, 1, 3, 3, 21, 21]$, duality $[0, 2, 1, 4, 3, 6, 5]$, fusion data:

1 0 0 0 0 0 0 0	0 1 0 0 0 0 0 0	0 0 1 0 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	1
0 1 0 0 0 0 0 0	0 0 1 0 0 0 0 0	1 0 0 0 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	0
0 0 1 0 0 0 0 0	1 0 0 0 0 0 0 0	0 1 0 0 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	1
0 0 0 0 1 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 1 2 0 0	1 1 1 1 1 0 0 0	1 1 1 1 1 0 0 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	3
0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	1 1 1 1 1 0 0 0	0 0 0 0 2 1 0 0	0 0 0 0 2 1 0 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	3
0 0 0 0 0 0 1 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 10 10	1 1 1 3 3 0 10	10
0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	1 1 1 3 3 0 10	10 10 10 10	0 0 0 0 0 0 11 10	10

- Formal codegrees: $[3, 3, 7, 7, 43, 43, 903]$,
- Property: extension of $\text{ch}(C_7 \rtimes C_3)$,
- Categorification status: excluded by [1, §6.4].

II.3. Rank 9. Among the MNSD Drinfeld rings of rank 9, there are 10 that are 1-Frobenius, and 2 that are neither perfect nor 1-Frobenius. Copy-pastable data can be found in the files `MNSD1FrobRank9DataOnly.txt` and `N1FrobMNSDRank9.txt`, located in the `Data/odd` directory of [8].

II.3.1. *1-Frobenius case.* Here is the complete list of 10 MNSD integral 1-Frobenius Drinfeld ring of rank 9:

- (1) FPdim 9, type $[1, 1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 8, 7, 6, 5, 4, 3, 2, 1]$, fusion data:

001000000000	010000000000	001000000000	000010000000	000001000000	000000100000	000000010000	000000001000	000000000100	000000000001
001000000000	000000000001	000000000100	000000000100	000000000100	000010000000	000010000000	000000010000	000001000000	100000000000
001000000000	000000000100	000000000100	000000000001	010000000000	010000000000	010000000000	000001000000	100000000000	000001000000
000010000000	000000000100	000000000001	000000000100	000000000100	000010000000	010000000000	100000000000	000001000000	000001000000
000001000000	000010000000	010000000000	001000000000	000000010000	100000000000	100000000000	000000000001	000000000100	000000000100
000000000100	000001000000	000001000000	100000000000	000000000001	000000000001	000000000001	000100000000	010000000000	001000000000
000000000100	000001000000	100000000000	000000000000	000000000100	000000000001	010000000000	001000000000	001000000000	000100000000
000000000001	100000000000	000000000000	000001000000	000000000100	000000000100	000000000100	001000000000	000100000000	010000000000

- Formal codegrees: $[9, 9, 9, 9, 9, 9, 9, 9, 9]$,
- Property: pointed
- Categorification status: $\text{Vec}(C_3^2)$.

- (2) FPdim 9, type $[1, 1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 8, 7, 6, 5, 4, 3, 2, 1]$, fusion data:

[illegible]

- Formal codegrees: $[9, 9, 9, 9, 9, 9, 9, 9, 9]$,
- Property: pointed,
- Categorification status: $\text{Vec}(C_9)$.

- (3) FPdim 57, type $[1, 1, 1, 3, 3, 3, 3, 3, 3]$, duality $[0, 2, 1, 8, 7, 6, 5, 4, 3]$, fusion data:

001000000000	001000000000	001000000000	000010000000	000001000000	000000100000	000000001000	000000000100	000000000010	000000000001
001000000000	001000000000	100000000000	000010000000	000001000000	000000100000	000000001000	000000000100	000000000010	000000000001
001000000000	100000000000	010000000000	000010000000	000001000000	000000100000	000000001000	000000000100	000000000010	000000000001
000100000000	000100000000	000010000000	000000000012	00000100101	000011000100	000011000100	000011000100	0000001110	111001100
000001000000	000001000000	000001000000	00000010101	00000001020	000001001001	00001001001	000010001110	11110000001	0000011100
000000100000	000000100000	000000100000	00000110010	00000101001	00001000200	11101000100	000001100010	000001100010	0000000111
000000000000	000000000000	000000000000	000000000000	000000000000	000000000000	000000000000	000000000000	000000000000	000000000000
000000000010	000000000010	000000000010	000000001110	11110000001	0000011001	00001000101	00000201000	0000101010	0000101010
000000000001	000000000001	000000000001	11100011100	00000011100	00000000111	00000100011	00001010010	00001010010	000210000

- Formal codegrees: $[3, 3, 19, 19, 19, 19, 19, 19, 57]$,
- Categorification status: $\text{Rep}(C_{19} \rtimes C_3)$.

- Categorification status: excluded by [4, Theorem 5.1].

- Formal codegrees: $[3_2, 8, 8, 24, 24]$,
- Property: noncommutative,
- Categorification status: $\mathcal{C}(S_4, 1, A_4, 1)$, $\mathcal{C}(S_4, 1, S_3, 1)$.

[illegible]

- (8) FPdim 78, type $[1, 1, 1, 1, 1, 6, 6]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- (9) FPdim 96, type $[1, 1, 1, 3, 4, 4, 4, 6]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- (10) FPdim 168, type $[1, 1, 1, 2, 2, 2, 3, 12]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- (11) FPdim 168, type $[1, 1, 1, 3, 3, 7, 7, 7]$, duality $[0, 2, 1, 4, 3, 5, 6, 7]$, fusion data:

[illegible]

- (12) FPdim 240, type $[1, 1, 1, 1, 1, 1, 3, 15]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- (13) FPdim 240, type $[1, 1, 1, 3, 6, 8, 8, 8]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 15, 61, 3660]$,
- Property: noncommutative,
- Categorification status:

